

(C++) Computer Programming

Assignment - 3

1. For each of the following programs, point out the error, if any. If there is no error, determine the output of the programs.

(a)

```
int x = 0;

while (x < 4)
{
    if (x++ = 1)
        cout << "A";
    else
        cout < "B";
}
```

(b)

```
int x = 5;

while (x > 0)
{
    if (x % 2 = 0)
        cout << x;

    x--;
}
```

(c)

```
int x == 0;

while (x < 3)
{
    if (x++ || ++x)
        cout << x;

    x++;
}
```

(d)

```
int x = 0;

while (++x < 3)
{
    cout << x << "-";
}
```

(e)

```
int x = 1;

while (x <= 5)
{
    cout << x << "-";
    x--;
}
```

(f)

```
int x = 1;

while (x <= 5);
{
    cout << x << "-";
    x++;
}
```

(g)

```
int x = 0;

while (x < 5)
{
    if (!x)
        cout << "A";
    else
        cout << x;

    x += 2;
}
```

(h)

```
int a = 10, b = 20;

if (a > 5)
    if (b < 10)
        cout << "A";
    else
        cout << "B";
```

2. Write a program to print the following:
1 2 3 4 5 6 7 8 9 10 9 8 7 6 5 4 3 2 1
3. Write a program to find the sum of squares up to n , where n is an integer entered by the user through the keyboard.
4. Write a program to enter numbers until the user wants to stop. At the end, display the count of positive, negative, and zero values entered.
5. Write a program that takes an integer n from the user and prints all numbers from 1 to n , but skips printing multiples of 4.
6. Write a program to count the number of digits in a given integer.
7. Write a program to count how many times a given digit occurs in a number. The input should consist of the number and the digit whose occurrence is to be counted.
8. Write a program to compute n th fibonacci number.
9. Write a program to find all the prime numbers from the first n natural numbers, where n is the input of the program.

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